

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459426.8424 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.062
Transit depth (Rp/Rs)^2	1.92 +/- 1.72 [%]
Orbital Inclination (inc)	83.64 +/- 1.84
Airmass coefficient 1 (a1)	106.15 +/- 111.41
Airmass coefficient 2 (a2)	-3.0 +/- 1.38
Scatter in the residuals of the lightcurve fit is	393.75 %
Best Comparison Star	#1 - [302, 370]
Optimal Aperture	6.31
Optimal Annulus	9.3
Transit Duration (day)	0.021 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.062 vs 0.1592 (deviation 0.17σ)
Duration fit vs published	0.021 d vs 0.0946 d (deviation 1.29σ)
Mid-time O–C	4.6 min
Depth SNR	1.1
Residual scatter	393.75 %

Light curve

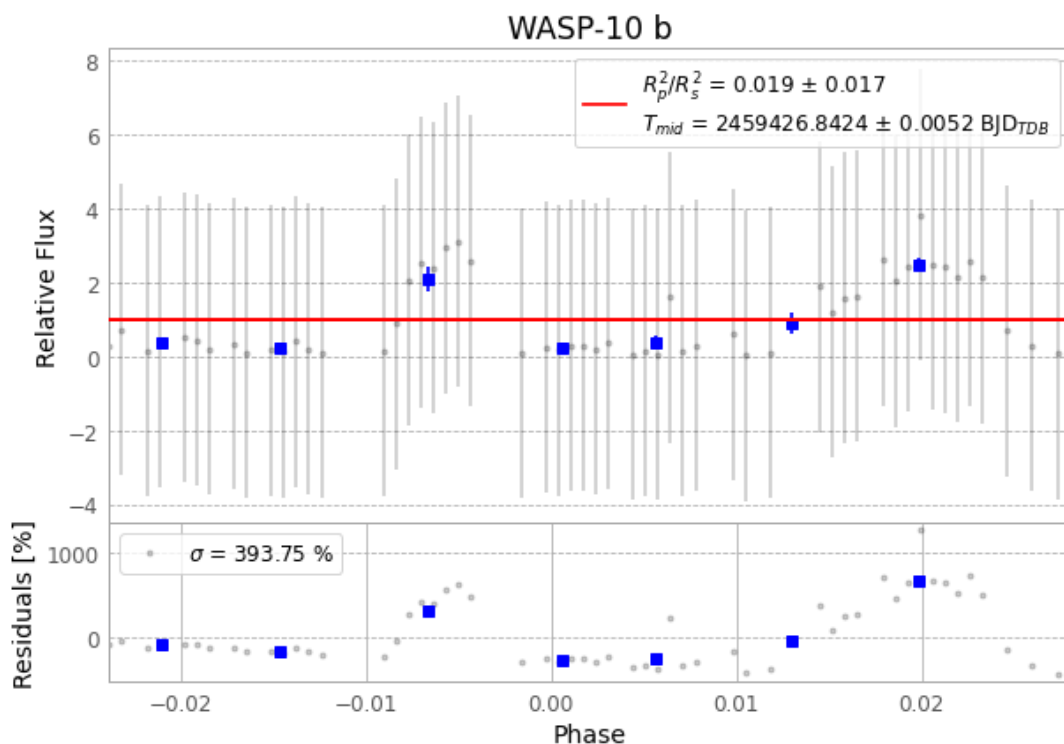


Figure 1 — FinalLightCurve_WASP-10 b_2021-07-30.png (SHA-256 ad7fb673c7f570b8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [596, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 08ce6e1463ff1be6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

80 FITS frames (53 MB), WASP-10210731061512.FITS → WASP-10210731101212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-210731.log (61a9af86a6a4...), FinalLightCurve_WASP-10 b_2021-07-30.png (ad7fb673c7f5...), FinalLightCurve_WASP-10 b_2021-07-30.csv (79a0a3ec2998...)

Compute resources

Wall time 295.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 01:54Z.